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Review Article

Electrochemical interface modelling for electrocatalytic materials design

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The advancement of net-zero emissions technologies requires an in-depth understanding of electrochemical reactions at electrified interfaces. Essential processes such as green hydrogen production and CO_2 reduction require sustainable electrocatalysts tailored for varied operational conditions. Computational techniques in electrocatalysis serve as crucial tools for providing microscopic insights and guiding towards higher-performing materials. Traditional modelling frameworks require approximations such as simplified surface models and an implicit description or neglect of electrolyte effects. A significant area for improvement is the treatment of the solid–liquid interface, where an explicit description of the electrolyte under realistic constant potential conditions remains the ultimate goal. This perspective examines recent advancements in charged interface modelling. We highlight cutting-edge simulation approaches, including the integration of machine learning techniques towards realistic atomic scale modelling for electrocatalytic materials design. As a case study, we focus on progress in understanding electrochemical nitrogen reduction for green ammonia production.

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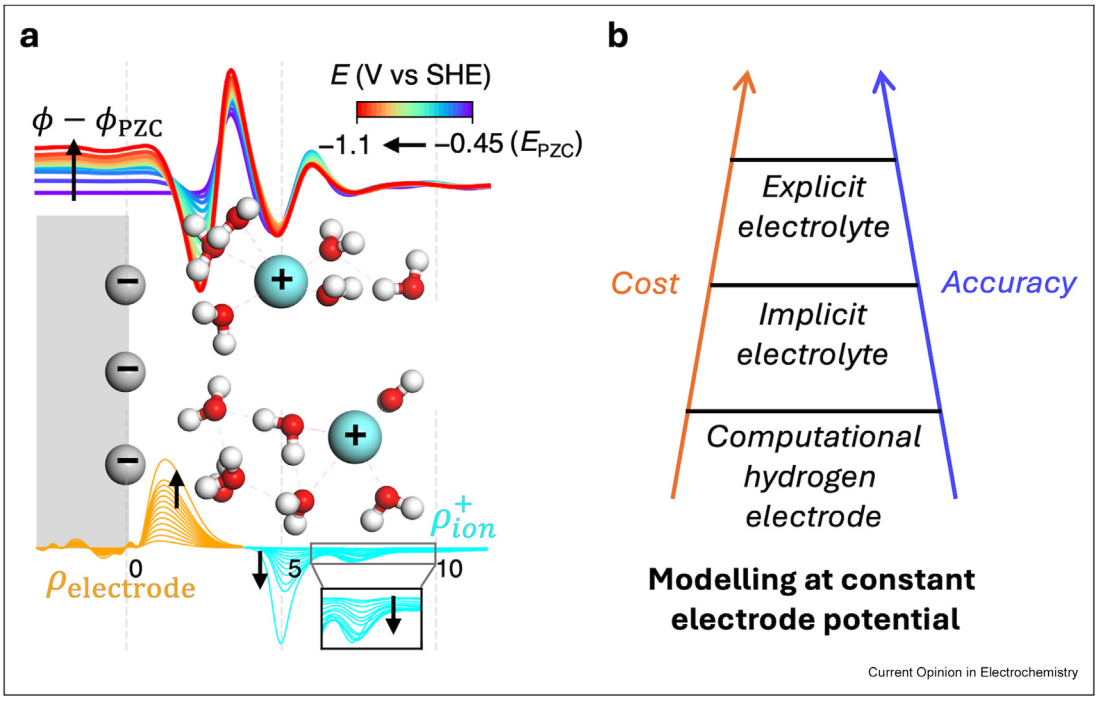
Introduction

The study of chemical reactions on electrified surfaces is crucial for the development of technologies necessary for a net zero emissions future [1,2]. For most technologies, such as green hydrogen production, CO_2 reduction to value-added products, and nitrogen fixation, there is an ongoing need to discover new materials to work as efficient electrocatalysts for anodic and cathodic reactions under different operational conditions. Computational electrocatalysis is an important tool to accelerate this process and has been successfully applied to explain experimental observations and trends [3–7], as well as to propose new materials [8–12] for such reactions.

The typical modelling framework within this field relies on density functional theory (DFT) calculations to predict the total energy and electronic structure combined with a set of approximations to relate these to electrochemical processes. These include: i) simplified surface models to represent the complex, and sometimes dynamic, character of the solid electrocatalysts; ii) reaction mechanisms assumed as sequential proton-coupled-electron transfer (PCET) steps and considered to be universal across different materials; iii) the computational hydrogen electrode (CHE) model [3] to account for the effect of changing the electrode potential; iv) contributions from the electrolyte are either ignored or assumed as a constant correction for all materials; v) neglect of kinetic parameters to focus on reaction thermodynamics; vi) a subset of reaction intermediates is considered due to the assumption of linear-scaling relationships.

Despite the success of the established computational framework, challenging the underlying approximations to provide a better description of the solid–liquid interface (SLI) is a key step for improving our understanding of the phenomena and prediction power for new materials. In this perspective, we address developments of constant potential calculations and SLI treatment using both implicit and explicit solvation methods (Figure 1), analyze the impact of these strategies for the case study of electrochemical nitrogen fixation for green ammonia production, and discuss

Figure 1



(a) Schematic solid–liquid interface structure. Electrostatic potential profile, ϕ, and electrode charge density, ρ, are shown with respect to the change of electrode potential, E. The values are subtracted from the potential at the point of zero charge, E_{PZC}. **(b)** Jacob's ladder of constant potential methods.

future challenges for the ongoing efforts to improve the design strategies for new electrocatalysts.

Implicit solvation and constant electrode potential

The coupling of implicit solvation methods with electronic structure techniques to simulate a constant electrode potential can include crucial aspects of the SLI for electrocatalysis in the modelling framework without a significant increase in the computational cost as compared to *in vacuo* models. Given the various options for implicit solvation models and distinct implementations to maintain constant electrode potentials, we refer the reader to focused reviews with in-depth discussions of each topic [13–19]. Here, we restrain ourselves to core concepts necessary to understand these approaches and use recent implementations to illustrate how the concepts are applied in available codes.

In the implicit solvation treatment, the solvent is described as a continuum dielectric medium that can screen the Coulomb potential between charged particles. The ability to screen this electrostatic interaction can be represented by a dielectric response function in the form of a damped Coulomb potential. Another aspect of implicit solvation models is the definition of a solvation cavity, that contains the solute (catalyst of interest) and is treated explicitly using quantum chemistry.

As we assume the existence of the two regions (inside and outside the solvation cavity), the system becomes a self-consistent reaction field problem. In this scenario, the electrolyte's charge density can be polarized by the solute's charge distribution, and its polarization can in turn induce new charge distributions within the solute. This problem, when including some concentration of dissolved ions, can be described via the Poisson-Boltzmann equation:

$$\nabla.[\varepsilon(\boldsymbol{r})\nabla \mathrm{v}(\boldsymbol{r})] = -4\pi(n_{solute}(\boldsymbol{r}) + n_{ions}(\boldsymbol{r}))$$

where $\varepsilon(\boldsymbol{r})$ is the dielectric response function that is equal to the vacuum permittivity within the cavity, adopts the value of the dielectric permittivity of the media far from the solute, and can be described by different approximations for the transition between the two values, $\mathrm{v}(\boldsymbol{r})$ is the electrostatic potential, $n_{solute}(\boldsymbol{r})$ is the charge density of the solute, and $n_{ions}(\boldsymbol{r})$ the ionic charge density that is given by the spatially dependent ionic distribution that depends on the electrostatic potential.

The various flavors of implicit solvation models can be distinguished by the definition of solvation cavity and dielectric response function [19–31], as well as the treatment for the ionic concentration [14,32–34]. For the solvation cavity, common approaches are the generation of cavities based on atom-centered spheres and radial cutoffs [20–23], directly based on isosurfaces of

the solute electronic density [24–27], or adaptations of these methodologies that can reduce solvent leakage artefacts. A tailored cavity construction allows the solvent description into regions near the solute that could not fit a solvent molecule or ion [28,29,31]. Meanwhile, the ionic concentration is commonly represented via a Boltzmann distribution or through modified-Boltzmann-Poisson (mPB) approaches designed to take ion size into account when determining its concentration in space [14,32–34].

The simulation of constant potential conditions together with implicit solvation methods can be achieved by leveraging DFT implementations in which the concentration of electrons adapts to ensure that the Fermi level of the electrode model matches the electrode potential, U [29,35–39], requiring the Helmholtz free energy to be evaluated as:

$$F(U) = E(U) - q_{solute}(U).U$$

where E(U) is the Helmholtz free energy and $q_{solute}(U)$ is the solute charge that depends on the applied potential. As we allow the variation in the number of electrons and make the periodic system charged, the electrostatic energy of the periodic cell will diverge. Modern constant potential implementations circumvent this problem by leveraging the implicit solvation models to accommodate the additional charge through ionic distributions during the solution of the mPB equations.

While the concepts discussed so far cover some foundational aspects, it is important to emphasize the existence of different implementation choices available. For example, Melander et al. point to the usage of the electrostatic potential averaged for a plane in the center of the electrode system and perpendicular to the surface normal, rather than the system's Fermi level, as the key quantity to be controlled [39]. Similarly, for charge neutralization, one can find examples of mPB equations coupled with Lagrange multipliers, background charges, as well as implementations that resort to linearized PB equations to ensure the neutrality of the cell [38,40–42]. Currently, implementations to explore constant potential coupled with implicit solvation can be found and used with different DFT codes, including VASP [29], JDFTx [35], ONETEP [36], QE [37], and GPAW [38,39].

The lower computational cost of implicit solvation schemes as compared to explicit solvation techniques is a strong advantage for the increasing popularity of such methods. However, care must be taken to avoid or mitigate common challenges that can arise with such approaches [43], making benchmark studies in this field extremely valuable for the community [43–46]. For instance, implicit solvation schemes have been shown to obtain surface diagrams in good agreement to the ones observed with explicit solvent depending on the chosen parameters for the solvation model [44]. Meanwhile, solvation energies of *CO, *CHO, *COH, *OCCHO, *OH, and *OOH adsorbed on Cu, Pt, and Au surfaces obtained through implicit solvation schemes were incapable of reproducing results from explicit calculations [45]. The same study also showed that the solvation energies of the adsorbates can differ for different metal surfaces, showing additional shortcomings of standard practice within the computational catalysis community. Similar conclusions were obtained when studying CO_2 adsorption on Fe/N_4C systems, in which conventional DFT simulations with implicit solvation models were not able to reproduce free energy changes observed with the inclusion of explicit solvation and constant potential considerations in the model [46].

Explicit solvation and constant electrode potential

Explicit solvation methods describe the SLI using molecular structures of the electrolyte. Molecular dynamics (MD) simulations are a primary tool for sampling the interfacial structure. The accuracy of this method depends on the level of theory employed and the simulation time accessible. The most widely used level is DFT with semi-local exchange-correlation functionals, such as PBE or RPBE [47,48]. This is often referred to as *ab initio* MD (AIMD). By increasing the level of theory, more detailed features can be investigated, such as using hybrid nonlocal functionals or path integral MD (PIMD) [49]. Conversely, lowering the level of theory can help to understand the system over a wider range of time and length scales, such as with QM/MM simulations [50]. Given the high computational cost associated with explicit solvation methods, the local concentration of electrolyte components at the interface may deviate from experimental values, which often converge over the length and time scales typically studied in AIMD simulations. This supports the rationale for multiscale approaches.

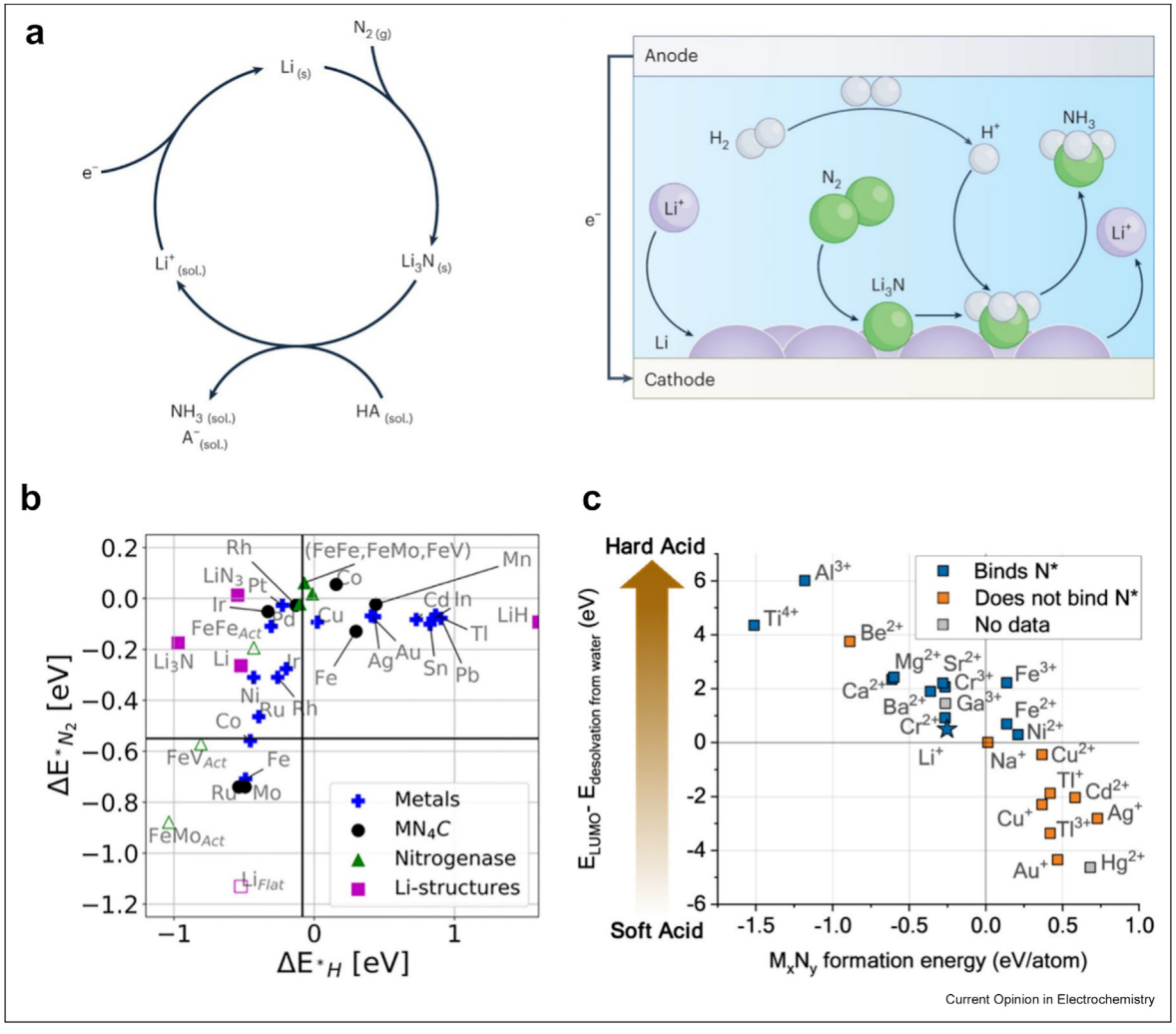
Most AIMD simulations of electrocatalysis employ the constant charge method rather than the constant potential method [47,48]. Grand-canonical descriptions of the interface with all explicit atoms are computationally demanding. While there are several AIMD studies that include explicit electrolytes at constant potential, they often also utilize implicit solvation methods [51–53]. By employing two electrodes with an electrolyte in between, the grand-canonical description of the system can be avoided, allowing for constant potential simulations with an explicit electrolyte at a reasonable computational cost [54]. In principle, this configuration is ideal for mirroring a real electrochemistry setup, but it also has several issues to overcome, e.g., limited length scale to achieve a bulk electrolyte regime; limited timescale to achieve steady-state conditions, and decoupling effects from the counter electrode [55].

Electrocatalytic interface modelling

Electrochemical N_2 reduction (N_2R) is a promising technology to replace the Haber-Bosch process which demands high pressure and temperature, but is still in its early stages [56]. One necessary objective for this application is to understand the reaction mechanism. Nitrogen dissociation via nitride formation sounds reasonable but a complex interface structure makes this not so straightforward [57]. Like many battery applications, at this SLI, an ill-defined multi-component disordered layer is generated under operating conditions [58]. Such interfacial effects (Figure 2a) lead to inconsistent results between different experimental groups [57,59] and challenges some of the common approximations in computational electrocatalysis.

The standard computational framework based on the CHE model, previously described in this perspective, has also been applied as a tool to select catalyst candidates for N_2R [4,60–62] and support experimental findings [63–65]. For example, early studies considering different metal surfaces indicated that the linear scaling relationship between the reaction intermediates would lead to a classical "volcano plot" and that all candidates could suffer from selectivity issues against the hydrogen evolution reaction [4,60]. The same methodology also pointed to other materials such as metal nitrides [61] and dual atom catalysts as possible candidates [62]. However, upon experimental verification, using more rigorous protocols to minimize false positives [66–70], the catalyst candidates proposed via DFT or tested by

Figure 2



(a) Illustration of a proposed N_2R reaction for Li catalysts, showing several complex interfacial processes that can be involved in the reaction [58]. **(b)** Classification scheme for electrocatalysts based on $^*N_2$ and H^* binding energies, Reprinted with permission from [76], Copyright 2021 American Chemical Society **(c)** Selection of catalyst candidates using $^*N$ binding energies, acidity descriptors, and nitride formation energies as figures of merit, Reproduced from [57].

experiments and supported via DFT have been shown to yield only background levels of ammonia.

The same experimental protocols [66] suggested Li-mediated NH_3 synthesis [71] as a process able to selectively produce NH_3 at promising rates. Calculations using similar assumptions as the ones from the CHE model have shown that lithium, lithium nitride, and lithium hydride surfaces could have limiting accessible potentials for N_2R [72]. However, the reasons behind the importance of the ill-defined layer and the bulk electrolyte for this reaction are still under debate [58,67,73–75] and will require more advanced computational methods to tackle. Similar complexity is found for other reactions, such as CO_2 reduction. Despite these difficulties, standard computational techniques can still be employed to search for options with analogous chemistries to Li (Figure 2b and c), providing guidance to further experimental efforts [57,76].

A deeper understanding of the reaction mechanism for N_2R would support catalyst optimization and discovery. In principle, details of such a complex interface can be dealt with constant potential approach with implicit and explicit solvation. However, considering different time or length scales for the formation of these ill-defined interphases, brute force simulation may not be successful. For instance, different electrolyte compositions can lead to distinct ill-defined layers [67,77] with thickness in the μm size range [77]. Modelling the ill-defined layer with appropriate SLI in the simulation cell and various computational modelling techniques could be a step forward to address this problem. Future efforts could also leverage from various computational modelling techniques to sample the potential energy surface of the proposed interface and create more realistic systems, including defects and distinct domains of the interface, that could be employed within kinetic models used to bring insights into Li-mediated N_2R [78,79].

Outlook and challenges

There are several obstacles to the widespread adoption of advanced computational electrochemical frameworks that represent the SLI and constant potential conditions more accurately. The first is the development of a standard procedure with community consensus. The CHE method can be applied based on vacuum calculations that are accessible to virtually all DFT codes. In contrast, methods for constant potential simulations with implicit or explicit solvation have distinct implementation choices and variables that make it difficult to compare results obtained with different approaches, are still being studied for different classes of materials [80], and present a barrier for new users. Among other initiatives, the development of shareable, accessible, and reproducible frameworks usable with different DFT implementations through scripting interfaces such as ASE could facilitate the widespread adoption of such approaches.

The second is the computational cost, with the increased length and timescales necessary to achieve a converged description of steady-state behaviour under an applied potential. This is also represented in the growing number of electrochemical datasets that can be used to support catalysts screening and design. For example, recent databases focused on CHE for catalyst screening can have nearly 1.2M DFT relaxations for metallic systems [81] and 62,000 for metal oxides [82]. In comparison, the recent BEASTdb database for implicit solvation and constant potential calculations [83], at this moment spans 20,000 surface calculations. Meanwhile, to the best of our knowledge, there are still no robust and dedicated databases systematically exploring AIMD or QM/MM explicit solvation techniques covering multiple catalyst candidates. The development and augmentation of tailored datasets with beyond CHE approaches will also contribute to more direct comparisons between distinct techniques, allowing consensus within the community to be reached.

One promising direction is the integration of machine learning force fields (MLFF). In contrast to classical force fields, which have a fixed functional form, MLFFs have a flexible form that is trained to reference data. The first generation of open-source pre-trained force fields can describe the energy of diverse chemical systems based on message-passing graph neural networks [84–86]. Recent and ongoing advances include the description of bond breaking/forming, redox processes, and the inclusion of long-range electrostatics. These are the essential ingredients required for electrochemical simulations. Although there are still further developments required, for instance, the accuracy level for complex cases such as the SLI has not been rigorously evaluated, and the constant potential ensemble has not been demonstrated, data-driven models have the potential to make advanced electrochemical simulations more accessible.

In summary, atomic scale simulations have been successfully developed and applied to understand electrocatalyst principles and processes. There is still room to make the computational techniques more realistic by removing approximations that limit the comparison to real electrochemical systems and experiments. For electrocatalysis, one example is our lack of understanding of the explicit solvent including the pH effect, ion-species effect, and additives effect. Incorporating a realistic solid catalyst surface beyond pristine low-index terminations is also a complicated problem. Morphological evolution of the catalyst during the reaction is one example. This includes not only producing various defects, e.g., point defects, and grain boundaries, but also generating an interphase that may play an active role in directing the catalytic pathway, or transition of precatalyst to catalyst (e.g from a metal to a metal hydroxide). Ultimately, such details could be incorporated

into theoretical kinetic models to provide quantitative comparison to experimental measurements and enable predictive guidance to new electrocatalytic systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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